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United States Patent	12399299
Kind Code	B2
Date of Patent	August 26, 2025
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Meta illuminator

Abstract

Provided are meta illuminators. The meta illuminators according to embodiments include a first light emitter configured to emit pattern light, and a second light emitter configured to emit non-patterned light, wherein the first and second light emitters forms a single body. The first and second light emitters respectively include meta-surfaces that are different from each other, and the different meta-surfaces may be formed on a single material layer. The first light emitter includes a pattern region that transmits a portion of incident light, and the second light emitter does not include the pattern region. A mask may be arranged between the light source and the transparent substrate.

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Appl. No.: 17/751259

Filed: May 23, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220350050 A1	Nov. 03, 2022

Foreign Application Priority Data

KR	10-2018-0077895	Jul. 04, 2018
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Related U.S. Application Data

Publication Classification

Int. Cl.: **G02B1/00** (20060101); **G02B27/42** (20060101); **H01S5/00** (20060101)

U.S. Cl.:

CPC **G02B1/002** (20130101); **G02B27/4233** (20130101); **G02B27/4272** (20130101);
G02B2207/101 (20130101); **H01S5/005** (20130101)

Field of Classification Search

CPC: G02B (1/002); G02B (27/4233); G02B (27/4272); G02B (27/425); G02B (27/4255);
G02B (2207/101); G02B (5/18); G02B (5/1804); G02B (5/1809); G02B (5/1814); G02B
(5/1819); G02B (5/1823); G02B (5/1842); G02B (5/1866); G02B (5/1871); G02B
(5/1876); H01S (5/005)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/199,759, filed Nov. 26, 2018, which claims the benefit of U.S. Provisional Application No. 62/626,263, filed on Feb. 5, 2018, in the US Patent Office and claims priority from Korean Patent Application No. 10-2018-0077895, filed on Jul. 4, 2018, in the Korean Intellectual Property Office, the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND

1. Field

(1) Apparatuses and methods consistent with exemplary embodiments relate to illuminators, and more particularly, to composite illuminators (hereinafter, meta illuminators) using a meta-surface based on a laser diode.

2. Description of the Related Art

(2) Illuminations may be used in a various applications besides the use of lighting a dark place. The use of illumination may be different according to a band of light. Light in an infrared band may be used in an optical sensor like a proximity sensor in a portable device, such as a mobile phone, and also, may be used as a two-dimensional (2D) illumination and a three-dimensional (3D) illumination for face recognition. The illumination may include non-coherent light or coherence light, and the light may be generated from a specific light source or an illumination device. Accordingly, an electronic device that uses light may include various illumination devices to obtain a desired illumination.

SUMMARY

(3) One or more exemplary embodiments provide meta illuminators having a reduced area in a device to which the meta illuminators are applied and a simplified arrangement.

(4) According to an aspect of an exemplary embodiment, meta illuminator includes: a first light emitter configured to emit pattern light; and a second light emitter configured to emit non-patterned light, wherein the first and second light emitters form a single body.

(5) In the meta illuminator, the first and second light emitters may include meta-surfaces that are different from each other, and the meta-surfaces different from each other may be formed on a single material layer.

(6) The first light emitter may include a pattern region that transmits a portion of incident light, and the second light emitter may not include the pattern region.

(7) The first and second light emitters respectively may include a light source, a transparent substrate, and a meta material layer that are sequentially stacked in the single body.

- (8) The meta illuminator may further include a single layer mask disposed between the light source and the transparent substrate.
- (9) The light source may be a light source emitting light in a direction perpendicular to the transparent substrate, and may include a first light source included in the first light emitter and a second light source included in the second light emitter.
- (10) The light source may be a light source emitting light in a direction parallel to the transparent substrate, and may include a first light source included in the first light emitter and a second light source included in the second light emitter.
- (11) The meta material layer may include a first meta-surface included in the first light emitter and a second meta-surface included in the second light emitter, wherein the first and second meta-surfaces have light emission characteristics different from each other and each of the first and second meta-surfaces includes a plurality of nano-posts.
- (12) The mask may include a pattern region in the first light emitter and a through-hole in the second light emitter. The pattern region is configured to generate structured pattern light and may include a meta-surface.
- (13) The meta illuminator may further include an element for changing a travelling direction of light emitted from the light source. As an example, the element may be a mirror or a prism.
- (14) The first meta-surface may be provided to emit structured pattern light. The first meta-surface may be a meta-diffractive optical element (DOE).
- (15) The second metal surface may be a micro-meta lens array arranged to emit uniform illumination.
- (16) According to another exemplary embodiment, the second meta-surface may be a DOE provided to emit uniform illumination.
- (17) The first and second light sources may be vertical cavity surface emitting lasers (VCSELs).
- (18) The first and second light sources may be edge emitting laser diodes.
- (19) According to an aspect of an exemplary embodiment, there is provided a meta illuminator including: a first light source and a second light source; a transparent substrate including: a upper surface on which a first meta-surface and a second-meta surface are disposed; and a lower surface on which a patterned mask is disposed and which faces the first light source and the second light source; wherein the meta illuminator comprises a first region and a second region which are separately provided and disposed side-by-side, and wherein the first light source, the transparent substrate, the first meta-surface, and the patterned mask are disposed in the first region so that a first light emitted from the first light source passes the patterned mask, the transparent substrate, and the first meta-surface, and wherein the second light source, the transparent substrate, the second meta-surface are disposed in the second region so that a second light emitted from the second light source passes the transparent substrate and the second meta-surface without passing through the patterned mask disposed in the first region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and/or other aspects will be more apparent by describing certain exemplary embodiments, with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a cross-sectional view of a first meta illuminator according to an exemplary embodiment;
- (3) FIG. 2 shows plan views of a first light source, a pattern region of a mask, and a first meta-surface that are included in a first part of a first meta illuminator of FIG. 1;
- (4) FIG. 3 is a 3D image of a nano-post array included in a first region of the first meta-surface of FIG. 2;

- (5) FIG. 4 is shows plan views of a second light source, a through-hole of a mask, and a second meta-surface that are included in a second part of a first meta illuminator of FIG. 1;
- (6) FIG. 5 is a plan view of a meta material layer of the first meta illuminator of FIG. 1;
- (7) FIG. 6 is a cross-sectional view of a second meta illuminator according to an exemplary embodiment;
- (8) FIG. 7 is a magnified cross-sectional view of a first region of a first meta-surface in the second meta illuminator of FIG. 6;
- (9) FIG. 8 is a magnified cross-sectional view of a second region of a second meta-surface in the second meta illuminator of FIG. 6;
- (10) FIG. 9 is a plan view of a meta material layer of the second meta illuminator of FIG. 6;
- (11) FIG. 10 is a 3D image of a plurality of nano-post array actually formed in a third region of a first meta region of FIG. 9;
- (12) FIG. 11 is a side-view of a third meta illuminator according to an exemplary embodiment;
- (13) FIG. 12 is a plan view showing a case that only a plurality of nano-posts that constitute a meta-surface are on an upper surface of a substrate in the meta illuminator of FIG. 11;
- (14) FIG. 13 is a plan view showing a light reflector and a light source supporter in a third meta illuminator of FIG. 11; and
- (15) FIG. 14 is a side-view of a fourth meta illuminator according to an exemplary embodiment.

DETAILED DESCRIPTION

- (16) Exemplary embodiments are described in greater detail below with reference to the accompanying drawings.
- (17) In the following description, like drawing reference numerals are used for like elements, even in different drawings. The matters defined in the description, such as detailed construction and elements, are provided to assist in a comprehensive understanding of the exemplary embodiments. However, it is apparent that the exemplary embodiments can be practiced without those specifically defined matters. Also, well-known functions or constructions are not described in detail since they would obscure the description with unnecessary detail.
- (18) It will be understood that, although the terms “first”, “second”, etc. may be used herein to describe various elements, the elements should not be limited by these terms. These terms are only used to distinguish one element from another element.
- (19) As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise. Also, it should be understood that, when a part “comprises” or “includes” a constituent element in the specification, unless otherwise defined, it is not excluding other elements but may further include other elements.
- (20) Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, the expression, “at least one of a, b, and c,” should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or any variations of the aforementioned examples.
- (21) FIG. 1 shows a composite illuminator **100** (hereinafter, referred to as the first meta illuminator) using a meta-surface according to an exemplary embodiment.
- (22) Referring to FIG. 1, the first meta illuminator **100** may include a light source supporter **10**, a substrate **16**, first and second light sources **12** and **14**, a mask **18**, and a meta-material layer **20**. The light source supporter **10** and the substrate **16** are sequentially stacked. The light source supporter **10** and the substrate **16** are separated from each other. The first and second light sources **12** and **14** are disposed on a surface, for example, an upper surface of the light source supporter **10**. The first and second light sources **12** and **14** are spaced apart by a given distance. The first and second light sources **12** and **14** may be the same or different light sources from each other. The first and second light sources **12** and **14** may be light sources that emit laser light. The first and second light sources **12** and **14** may be light sources that emit light towards the substrate **16** in direction perpendicular to the upper surface of the light source supporter **10**. The first and second light sources **12** and **14** may

be, for example, a vertical cavity surface emitting laser (VCSEL). Light may be directly irradiated to the substrate **16** by the first and second light sources **12** and **14**. The first and second light sources **12** and **14** may be, for example, light sources that emit light in an infrared region, but may not be limited thereto. The substrate **16** may be a transparent substrate with respect to light. The substrate **16** may be, for example, a silicon substrate. The meta material layer **20** is formed on an upper surface of the substrate **16**. Here, the meta material layer **20** may denote a material layer including a plurality of nano-posts that constitute a meta-surface. The meta material layer **20** may include nano-gratings, nano-holes, or nano-slits instead of the nano-posts. Also, the meta-material layer **20** may include at least one of the nano-posts, the nano-gratings, the nano-holes, and the nano-slits or may include a combination of two or more of the nano-posts, the nano-gratings, the nano-holes, and the nano-slits. The meta material layer **20** may include a first meta-surface **20A** and a second meta-surface **20B**. The first meta-surface **20A** may correspond to the first light source **12** and may be disposed above the first light source **12**. The second meta-surface **20B** may correspond to the second light source **14** and may be disposed above the second light source **14**. The first meta-surface **20A** may be a meta lens that projects a structured pattern light generated by a pattern region **18A** of the mask **18** to a given location. Accordingly, dimensions (width and height) of each of the first nano-posts **P1** that constitute the first meta-surface **20A** and an arrangement of the first nano-posts **P1** may be designed in consideration of functions of the first meta-surface **20A**. Heights and pitches of the first nano-posts **P1** may be the same and a width and a diameter of each of the first nano-posts **P1** may be different, but such conditions are not limited thereto. Each of the first nano-posts **P1** may have a size less than a wavelength of incident light. Light transmission characteristics, for example, a phase and/or intensity of each of the first nano-posts **P1** with respect to incident light may be controlled through controlling of the size and type of each of the first nano-posts **P1**. Accordingly, the light transmission characteristics of each of the first nano-posts **P1** may be designed according to locations of the first nano-posts **P1**. Accordingly, a wave front of incident light may be transformed to an arbitrary shape by appropriately arranging the first nano-posts **P1**. As an example, a lens that focuses or diffuses incident light may be realized by using a meta-surface.

(23) The second meta-surface **20B** may be configured to provide light for a two-dimensional (2D) image or a three-dimensional (3D) image. The second meta-surface **20B** may diffuse light incident to the second meta-surface **20B** to have a uniform intensity. Light of uniform intensity may be irradiated by the second meta-surface **20B** to an entire area of a target illumination region. In other words, the second meta-surface **20B** may be configured to obtain illumination with uniform intensity of light on the entire area of the target illumination region to be irradiated. The second meta-surface **20B** may include a plurality of micro-meta lenses **ML1**. Each of the micro-meta lenses **ML1** may have a size in a range of about a few μm to about a few hundreds of μm . The micro meta lenses **ML1** may have the same shape and size. The plurality of micro-meta lenses (**ML1**) may be arranged in a given shape. As an example, as depicted in FIG. 5, the plurality of micro meta lenses **ML1** may form an array (hereinafter, a micro-meta lens array) in which the micro meta lenses **ML1** are arranged in a horizontal direction and a vertical direction.

(24) Each of the micro meta lenses **ML1** that form the second meta-surface **20B** may perform a function of diverging incident light. Since the plurality of micro meta lenses **ML1** are the same, in a view of the whole second meta-surface **20B**, light of a uniform intensity may be emitted from the second meta-surface **20B**.

(25) Each of the plurality of micro meta lenses **ML1** that constitute the second meta-surface **20B** may be referred to as a unit micro meta lens constituting the micro meta lens array. The unit micro meta lens, like the first meta-surface **20A**, may include a plurality of nano-posts, and the dimension of each of the nano-posts and arrangement of the nano-posts may be designed considering the functions of the unit micro meta lens. In the micro meta lens array that constitutes the second meta-surface **20B**, the unit micro meta lenses may have the same size and the same function, and thus, a

height of the plurality of nano-posts included in the unit micro meta lenses may be the same, and also, pitches or intervals thereof may be the same.

(26) The dimensions (e.g., height, interval, width, etc.) of the first nano-posts **P1** of the first meta-surface **20A** and the dimensions of the nano-posts included in the micro meta lenses **ML1** that constitute the second meta-surface **20B** may be controlled in a manufacturing process depending on required functions of the first and second meta-surfaces **20A** and **20B**. That is, the light transmission characteristics or light transfer characteristics of the first nano-posts **P1** and the nano-posts included in the micro meta lenses **ML1** may be controlled in a manufacturing process. Thus, if necessary, the first and second meta-surfaces **20A** and **20B** may be designed to other optical elements (e.g., diffractive optical element, etc.).

(27) The first nano-posts **P1** and the nano-posts included in the micro meta lenses **ML1** may have a refractive index greater than that of a surrounding material by 1 or greater. For example, if the substrate **16** has a refractive index of 2, a material included in the first nano-posts **P1** and the nano-posts included in the micro meta lenses **ML1** may have a refractive index of 3 or more.

(28) The mask **18** is disposed between the first and second light sources **12** and **14**. According to an exemplary embodiment, the mask **18** may be attached to a lower surface of the substrate **16**. The mask **18** may be directly contact the lower surface of the substrate **16**. In some other exemplary embodiments, another transparent member may be disposed between the mask **18** and the substrate **16**. The mask **18** may be a metal mask. The mask **18** may include a pattern region **18A** and a through-hole **18h**. The pattern region **18A** corresponds to the first light source **12** and the first meta-surface **20A**. The through-hole **18h** corresponds to the second light source **14** and the second meta-surface **20B**. The pattern region **18A** may be a region where a plurality of patterns **18A1** are formed and separated from each other in the through-hole **18h**. The plurality of patterns **18A1** of the pattern region **18A** may be metal patterns. Accordingly, light incident to the plurality of patterns **18A1** of light incident to the pattern region **18A** is reflected, and light incident between the patterns **18A1** enters the substrate **16**, and then, enters the first meta-surface **20A**. Thus, a structured pattern light is generated by the pattern region **18A**. Accordingly, the plurality of patterns **18A1** may be designed so that the structured pattern light is generated.

(29) The plurality of patterns **18A1** of the pattern region **18A** may have the same height. Also, intervals between the plurality of patterns **18A1** may be equal to each other. However, a width of each of the plurality of patterns **18A1** may be different from each other. These conditions are not limited thereto, and may be changed depending on the function of the mask **18**. The through-hole **18h** may have a diameter that is greater than the size of the second light source **14**. The through-hole **18h** may be arranged so as not to interrupt light emitted from the second light source **14** to enter an entire region of the second meta-surface **20B**. That is, the through-hole **18h** may have a size so that light emitted from the second light source **14** is prevented from being incident to the first meta-surface **20A** while allowing the light emitted from the second light source **14** to be incident to the entire region of the second meta-surface **20B**.

(30) As an example, the pattern region **18A** of the mask **18** may be a meta-surface including a plurality of nano-posts. In this case, the arrangement of the plurality of nano-posts and each of the plurality of nano-posts may be designed to generate the structured pattern light. As an example, when the pattern region **18A** is a meta-surface, heights, widths, pitches, etc. of the plurality of nano-posts of the meta-surface may be determined to generate the structured pattern light in a manufacturing process.

(31) In another example, the entire mask **18** may be a meta-surface. At this point, the pattern region **18A** of the mask **18** may be designed to generate the structured pattern light, and the remaining part may be designed to reflect incident light.

(32) The first meta illuminator **100** may be divided into a first part **100A** configured to generate a structured pattern light and a second part **100B** configured to irradiate light of uniform intensity to the entire region to be irradiated. That is, the first meta illuminator **100** may include the first part

100A configured to emit a structured pattern light and a second part **100B** configured to irradiate light of uniform intensity to the entire region to be irradiated. The pattern region **18A** may exist on the substrate **16** so that the light emitted from the second light source **14** reaches the substrate **16** without passing through the pattern region **18A**. The division of the first meta illuminator **100** may be applied to second through fourth meta illuminators which will be described below.

(33) According to the descriptions above, the first part **100A** of the first meta illuminator **100** may be a part including the first light source **12**, the pattern region **18A** of the mask **18**, the substrate **16**, and the first meta-surface **20A**. The second part **100B** may be a part including the second light source **14**, the through-hole **18h** of the mask **18**, the substrate **16**, and the second meta-surface **20B**. Illuminations generated from the first part **100A** and the second part **100B** are different from each other, and thus, the first meta illuminator **100** may be referred to as a composite illuminator, and for the same reason, the second through fourth illuminators which will be described below may also be referred to as composite illuminators.

(34) Meanwhile, the first meta illuminator **100** may include a driving unit for driving the first and second light sources **12** and **14**. The first and second light sources **12** and **14** may be operated in a time-sharing method by the driving unit. Also, the first and second light sources **12** and **14** may be simultaneously operated.

(35) FIG. 2 shows plan views of the first light source **12**, the pattern region **18A** of the mask **18**, and the first meta-surface **20A** included in the first part **100A** of the first meta illuminator **100** of FIG. 1.

(36) In FIG. 2, reference numeral **40** denotes an optical axis.

(37) Referring to FIG. 2, the first light source **12** may include a plurality of unit light sources **12A** that constitute an array. In FIG. 2, it is depicted that the first light source **12** includes the four unit light sources **12A** for convenience. However, an array of the unit light sources **12A** may be formed by including more than four unit light sources **12A** to supply sufficient light to the first meta-surface **20A**. Each of the unit light source **12A** may be a VCSEL. It is seen that the pattern region **18A** of the mask **18** includes a hole **18Ah** passing through the mask **18** and a plurality of patterns **18A1** distributed in the hole **18Ah**. Each of the patterns **18A1** may be a nano-post. Widths of the patterns **18A1** may differ from each other or may be the same. The patterns **18A1** may be arranged to form a symmetrical shape or an asymmetrical shape. Also, the patterns **18A1** may be formed as repeated patterns. Also, the patterns **18A1** may be randomly distributed. That is, the patterns **18A1** may be distributed in an arbitrary form, and each of the patterns **18A1** may have an arbitrary shape. The hole **18Ah** of the pattern region **18A** is rectangular shape, but may be a circular shape or other shapes. The first meta-surface **20A** has an appearance of a circular shape, but is not limited thereto, that is, may have a rectangular shape or other shapes. The first meta-surface **20A** may have an area or a diameter that sufficiently accommodates all light incident through the pattern region **18A** of the mask **18**.

(38) FIG. 3 is a 3D image of a nano-post array included in a first region **20A1** of the first meta-surface **20A** of FIG. 2. An arbitrary nano-post in the nano-post array of FIG. 3 may correspond to the first nano-posts **P1** of the first meta-surface **20A** of FIG. 1.

(39) FIG. 4 is a plan view of the second light source **14**, the through-hole **18h** of the mask **18**, and the second meta-surface **20B** included in the second part **100B** of the first meta illuminator **100** of FIG. 1.

(40) In FIG. 4, reference numeral **50** denotes an optical axis.

(41) Referring to FIG. 4, the second light source **14** may include a plurality of unit light sources **14A** that constitute an array. In FIG. 4, it is depicted that the second light source **14** includes four unit light sources **14A** for convenience, but the second light source **14** may form an array by including more than four unit light sources **14A** to supply sufficient light to the second meta-surface **20B**. An area and diameter of the through-hole **18h** formed in the mask **18** may be greater than the second light source **14**. There is no pattern that may interfere with the travel of light in the

through-hole **18h**. Accordingly, light emitted from the second light source **14** is directly incident to the substrate **16**, and thus, may be incident to the second meta-surface **20B**. The second meta-surface **20B** may include the plurality of micro meta lenses **ML1** disposed in a through-hole **20h** formed in the meta material layer **20**. The micro meta lenses **ML1** may form an array. Each of the micro meta lenses **ML1** may have a circular shape, but is not limited thereto, that is, may have a rectangular shape or other shapes.

(42) FIG. 5 is a plan view of the meta material layer **20** of the first meta illuminator **100** of FIG. 1.

(43) Referring to FIG. 5, as described with reference to FIG. 2, an appearance of the first meta-surface **20A** has a circular shape. For convenience of explanation, the plurality of nano-posts that constitute the first meta-surface **20A** are not depicted. An overall appearance of the second meta-surface **20B** is a rectangular shape, but is not limited thereto, and may have a circular shape. The first meta-surface **20A** and the second meta-surface **20B** are separated from each other.

(44) FIG. 6 is a cross-sectional view of a composite illuminator **200** (hereinafter, a second meta illuminator) that uses a meta-surface according to an exemplary embodiment. Parts different from the first meta illuminator **100** will be described.

(45) Referring to FIG. 6, the second meta illuminator **200** may include the first and second light sources **12** and **14**, the transparent substrate **16**, and a meta material layer **30**. The first and second light sources **12** and **14** are mounted on a supporter **10**. Lights emitted from the first and second light sources **12** and **14** are directly incident to the transparent substrate **16**. That is, there is no other member between the transparent substrate **16** and the first and second light sources **12** and **14**. The meta material layer **30** is disposed on an upper surface of the transparent substrate **16**. The meta material layer **30** may directly contact the upper surface of the transparent substrate **16**. The meta material layer **30** may include a first meta-surface **32** and a second meta-surface **34**. The first meta-surface **32** corresponds to the first light source **12**. The second meta-surface **34** corresponds to the second light source **14**. The first and second meta-surfaces **32** and **34** are separated from each other in the meta material layer **30**. The first meta-surface **32** and the second meta-surface **34** may include a plurality of nano-posts. The first meta-surface **32** and the second meta-surface **34** may function as a diffractive optical element (DOE). Arrangements and shapes of each of the plurality of nano-posts included in the first meta-surface **32** may be designed to form a structured pattern light by diffracting incident light. The plurality of nano-posts included in the second meta-surface **34** may be designed so that light having uniform intensity is irradiated to an entire region to be irradiated by diffracting incident light. Accordingly, the first meta-surface **32** and the second meta-surface **34** may function as meta DOEs.

(46) FIG. 7 is a magnified cross-sectional view of a first region **A1** of the first meta-surface **32** of the second meta illuminator **200** of FIG. 6.

(47) Referring to FIG. 7, the first region **A1** may include first, second, and third nano-posts **32A**, **32B**, and **32C** on the substrate **16**. The first through third nano-posts **32A** through **32C** may have the same height, but may have widths different from each other. The shapes and arrangements of the first through third nano-posts **32A** through **32C** of FIG. 7 are arbitrary. The arrangements and configurations of the plurality of nano-posts of the first meta-surface **32** may be differently designed depending on a shape of a structured pattern light.

(48) FIG. 8 is a magnified cross-sectional view of a second region **A2** of the second meta-surface **34** in the second meta illuminator **200** of FIG. 6.

(49) Referring to FIG. 8, first, second, and third nano-posts **34A**, **34B**, and **34C** are formed on the substrate **16**. The first through third nano-posts **34A** through **34C** may have the same height. The first through third nano-posts **34A** through **34C** may have widths different from each other. The arrangements and shapes of the first through third nano-posts **34A** through **34C** on the substrate **16** are arbitrary. The arrangements and configurations of the plurality of nano-posts that constitute the second meta-surface **34** may be different depending on a required flood illumination.

(50) FIG. 9 is a plan view of the meta material layer **30** of the second meta illuminator **200** of FIG.

6.

- (51) Referring to FIG. 9, the meta material layer **30** may include the first and second meta-surfaces **32** and **34** spaced apart from each other. The first and second meta-surfaces **32** and **34** are a rectangular shape, but is not limited thereto. For example, in the case of the first meta-surface **32**, an appearance thereof may be a circular shape.
- (52) FIG. **10** is a 3D image of a plurality of nano-post arrays actually formed in a third region A3 of the first meta region **32** of FIG. **9**.
- (53) Referring to FIG. **10**, a first nano-post **60** in the plurality of nano-post arrays may correspond to one of the first through third nano-posts **32A** through **32C** formed on the first meta-surface **32** of FIG. **7**, and a second nano-post **62** may correspond to another one of the first through third nano-posts **32A** through **32C** formed on the first meta-surface **32** of FIG. **7**.
- (54) FIG. **11** shows a composite illuminator **300** (hereinafter, a third meta illuminator) using a meta-surface according to an exemplary embodiment.
- (55) Referring to FIG. **11**, the third meta illuminator **300** does not employ a direct light irradiation method in which light L1 emitted from a light source **72** is directly irradiated to a substrate **78**. third meta illuminator **300** employs an indirect light irradiation method in which light L1 emitted from the light source **72** is reflected by a light reflector **76**, and afterwards, enters the substrate **78**. In the third meta illuminator **300**, the light source **72** emits the light L1 in a direction parallel to the substrate **78**, that is, in a horizontal direction. The light source **72** may be an edge emitting laser diode or may include an edge emitting laser diode. Since this type of light source **72** is used, a thickness of the third meta illuminator **300** may be relatively thin. The light source **72** is mounted on a light supporter **70**. A sub-supporter may be arranged on the light supporter **70** and the light source **72** may be mounted on the sub-supporter. The light reflector **76** may be disposed in front of a light emitting surface of the light source **72**. Accordingly, the light L1 emitted from the light source **72** may firstly be incident to the light reflector **76**. The light reflector **76** is an example member of changing a travelling direction of the light L1. However, the light reflector **76** of FIG. **11** is not limited thereto. As an example, as depicted in FIG. **11**, the light reflector **76** may function as a mirror that reflects incident light to a direction different from an incident direction. As another example, the light reflector **76** may be replaced by a refractive member that refracts the incident light to different directions. For example, the light reflector **76** may be replaced by a prism.
- (56) Referring to FIG. **11**, the light reflector **76** includes a mirror surface S1. The mirror surface S1 may be an inclined surface with a given inclination angle (e.g., 45°) with respect to a bottom surface of the light reflector **76**. Since the mirror surface S1 has a given inclination angle, the light L1 incident to the mirror surface S1 from the light source **72** is reflected by the mirror surface S1, and thus, may enter the substrate **78** in a direction perpendicular to the bottom surface of the substrate **78**. The mirror surface S1 may be disposed on a front face of a light emission surface of the light source **72** or a part from which the light L1 is emitted. The substrate **78** is disposed above the light reflector **76** and the light source **72**. The substrate **78** is disposed to cover the light reflector **76** and the light source **72** with the mirror surface S1 of the light reflector **76** as the center. Since the light source **72** is mounted on the light supporter **70**, at least a part of the light supporter **70** may be covered by the substrate **78**. The substrate **78** may be disposed parallel to the light source **72**. A material layer **80** is formed on an upper surface of the substrate **78**. The material layer **80** may include a first meta-surface **80A**. The first meta-surface **80A** may function as a meta lens. A plan view of the material layer **80** may be the same as the plan view of the material layer **20** of FIG. **5**. An arrangement of patterns of the first meta-surface **80A** may be the same as the arrangement of the patterns that constitute the first meta-surface **20A** of FIG. **5**. The material layer **80** may include a second meta-surface that is the same as the second meta-surface **20B** of FIG. **5**. That is, in the third meta illuminator **300**, the material layer **80** including a meta-surface may include the same material and the meta-surface configuration of the material layer **20** of FIG. **5**. As another example, parts of the material layer **80** that surrounds the meta-surface may be removed. That is, only the

first meta-surface **80A** may remain on the upper surface of the substrate **78**. In this case, only the first meta-surface **80A** corresponding to the first meta-surface **20A** of FIG. 5 and another meta-surface corresponding to the second meta-surface **20B** may remain on the upper surface of the substrate **78**. For example, as depicted in FIG. 12, only first nano-posts **80A1** that constitute the first meta-surface **80A** and a plurality of micro meta lenses **80B1** that constitute the second meta-surface **80B** may remain on the upper surface of the substrate **78**.

(57) Referring to FIG. 11, a mask **82** is disposed on a bottom surface of the substrate **78**. The mask **82** may directly or indirectly contact the bottom surface of the substrate **78**. The mask **82** may be a metal mask. The mask **82** is separated from the light source **72** and the light reflector **76**. In FIG. 11, only a side of the mask **82** is shown, but the entire configuration of the mask **82** may be the same as the mask **18** described with reference to the first meta illuminator **100**. That is, the mask **82** includes a pattern region **82A** on a region corresponding to the first meta-surface **80A**. The pattern region **82A** may include a plurality of patterns **82p** formed in a through-hole **82h** of the mask **82**. The plurality of patterns **82p** are separated from each other and each may be a metal post. When the light **L1** enters the pattern regions **82A**, the light **L1** incident to the plurality of patterns **82p** is reflected, and the light **L1** incident between the plurality of patterns **82p** enters the substrate **78**, and then, enters the first meta-surface **80A** which is a meta lens arranged on the upper surface of the substrate **78**, through the substrate **78**. The plurality of patterns **82p** in the pattern region **82A** may be designed to generate a structured pattern light. The structured pattern light generated in the pattern region **82A** may be projected to a front of the upper surface of the substrate **78** by the first meta-surface **80A**.

(58) The mask **82** may be replaced by a meta-surface like the mask **18** of the first meta illuminator **100**.

(59) FIG. 13 is a plan view of the light reflector **76** and the light supporter **70** in the third meta illuminator **300** of FIG. 11.

(60) Referring to FIG. 13, first and second light sources **72** and **74** are disposed parallel to each other on the light supporter **70**. The first and second light sources **72** and **74** are separated from each other. The first light source **72** may be used to form a structured pattern light by irradiating light to the first meta-surface **80A**. The second light source **74** may be used to form uniform illumination for a 2D/3D image by irradiating light to the second meta-surface **80B**. The light reflector **76** is disposed in front of light-emitting surfaces of the first and second light sources **72** and **74**. In FIG. 13, arrows directed towards the light reflector **76** indicate laser beam emitted from the first and second light sources **72** and **74**. A mirror surface **S1** of the light reflector **76** faces the first and second light sources **72** and **74**. The mirror surface **S1** may have a given length in a direction perpendicular to the first and second light sources **72** and **74**. The mirror surface **S1** is disposed parallel to the light supporter **70** along the light supporter **70** from one end to the other end of the light supporter **70**. Accordingly, even though lights are simultaneously emitted from the first and second light sources **72** and **74**, the lights emitted from the first and second light sources **72** and **74** may be reflected towards the first and second meta-surfaces **80A** and **80B** by the light reflector **76**. In FIG. 13, the light reflector **76** may be replaced by a light refracting element. The light refracting element may be disposed to transmit incident light to the first and second meta-surfaces **80A** and **80B** by a refraction function of the light refracting element. The light refracting element may be, for example, a prism.

(61) FIG. 14 is a side-view of a composite illuminator **400** (hereinafter, a fourth meta illuminator **400**) using a meta-surface according to an exemplary embodiment. Parts of the fourth meta illuminator **400** different from the third meta illuminator **300** will be described.

(62) Referring to FIG. 14, a meta material layer **90** is formed on the upper surface of the substrate **78**. The meta material layer **90** may directly contact the upper surface of the substrate **78**. The meta material layer **90** may include a first meta-surface **90A**. The first meta-surface **90A** includes a plurality of nano-posts **90A1**. The plurality of nano-posts **90A1** may be arranged or designed so

that the first meta-surface **90A** functions as a DOE (hereinafter, a first meta DOE). Heights of the plurality of nano-posts **90A1** may be generally the same. Also, pitches between the plurality of nano-posts **90A1** may be the same or generally the same. However, widths of the plurality of nano-posts **90A1** may be different from each other. A planar configuration of the meta material layer **90** may be the same as that of the meta material layer **30** of FIG. **9**. That is, the meta material layer **90** may include the first meta-surface **90A** and another meta-surface corresponding to the second meta-surface **34** of FIG. **9**. The other meta-surface may also include a plurality of nano-posts like the first meta-surface **90A**. The plurality of nano-posts included in the other meta-surface may be disposed or designed to perform a function of a DOE (hereinafter, a second meta DOE). Light emitted from the first light source **72** may be generated as a structured pattern light by the first meta DOE. By the second meta DOE light emitted from the second light source **74** may be irradiated with uniform intensity on the entire region where the light is irradiated. The meta illuminators according to the exemplary embodiments described above are composite illuminators. However, they may be referred to as monolithic illuminators in that the illuminators that generate illumination having different characteristics from each other from a single unit layer.

(63) In the case of an existing mobile phone, two or more illuminations may be needed. For example, three illuminations (e.g., a proximity sensor, a 2D and 3D illumination for face recognition) may be provided on a front part of the mobile phone, and these three illuminators may be individually mounted on the mobile phone. Accordingly, in the case of the existing mobile phone, three independent mounting spaces may be needed for the three illuminators. The current trend of the existing mobile phone is that a display area on a front side of the mobile phone is increased. However, in the case of an existing mobile phone, since a plurality of illuminators or illumination parts are individually used, it is difficult to reduce an area occupied by the entire illumination device at the front part of the mobile phone. Moreover, in the case of an existing mobile phone, as the number of the required illuminators increase, an area occupied by all the illuminators is increased at the front part of the mobile phone. As a result, in the case of an existing mobile phone, a form-factor is increased and individual illuminators should be mounted one by one, and thus, this may be a cause of increasing manufacturing costs and the alignment in the manufacturing process may be complicated compared to the case of mounting a single illuminator.

(64) However, in the case of the illuminators according to the exemplary embodiments, each of the illuminators is a composite illuminator based on a laser diode and using a meta-surface, and different illuminations are emitted from a single illuminator. Accordingly, when the meta illuminators according to the exemplary embodiments are used, since only one illuminator is mounted, most of the limitations of an existing mobile phone may be removed.

(65) Meta illuminators or the concept of the meta illuminators according to the embodiments may be applied to not only a mobile phone, such as a smart phone, but also may be applied to various devices, for example, an illumination unit of a small portable device, such as watches, telescopes, or microscopes (glasses), etc. and also a structured light (SL) illumination projector for a depth sensor, or a compact laser light scanner.

(66) The foregoing exemplary embodiments are merely exemplary and are not to be construed as limiting. The present teaching can be readily applied to other types of apparatuses. Also, the description of the exemplary embodiments is intended to be illustrative, and not to limit the scope of the claims, and many alternatives, modifications, and variations will be apparent to those skilled in the art.

Claims

1. An illuminator comprising: a light source supporter; a substrate disposed to correspond to the light source supporter; a light source disposed on a side of the light source supporter; and a meta surface on a first surface of the substrate, wherein the light source and the meta surface are

disposed to correspond to each other, wherein the meta surface comprises a plurality of nano-structures and each of the plurality of nano-structures has dimension less than a wavelength of light incident on the first surface, wherein the light source is between the light source supporter and the substrate, and wherein the meta surface comprises: a first meta surface configured to project a patterned light on a given position; and a second meta surface configured to provide light for a two dimensional (2D) or three dimensional (3D) image.

2. The illuminator of claim 1, further comprising a mask pattern on a second surface of the substrate, wherein the mask pattern comprises patterns different from the meta surface.

3. The illuminator of claim 2, wherein the mask pattern comprises: a pattern region comprising a plurality of patterns; and a through-hole spaced apart from the pattern region, wherein the plurality of patterns are arranged to generate a patterned light.

4. The illuminator of claim 3, wherein the pattern region comprises meta patterns configured to generate the patterned light.

5. The illuminator of claim 1, wherein the light source comprises a plurality of light sources spaced apart from each other.

6. The illuminator of claim 1, wherein the first meta surface comprises a plurality of first nano-structures, wherein the second meta surface comprises a plurality of second nano-structures, and wherein the plurality of first nano-structures are arranged to form a patterned light by diffracting light incident on the first meta surface, and the plurality of second nano-structures are arranged to emit uniform illumination to a whole area to which light is illuminated by diffracting light incident on the second meta surface.

7. The illuminator of claim 6, wherein the first meta surface comprises a meta lens.

8. The illuminator of claim 6, wherein the first meta surface comprises a meta diffractive optical element (DOE).

9. The illuminator of claim 6, wherein the second meta surface comprises a meta diffractive optical element provided to emit a non-patterned light having uniform illumination.

10. The illuminator of claim 1, wherein the light source comprises a plurality of light sources disposed to emit light in a direction perpendicular to the substrate and spaced from each other.

11. The illuminator of claim 1, wherein the light source comprises a plurality of light sources disposed to emit light in a direction parallel to the substrate and spaced from each other.

12. The illuminator of claim 11, further comprising an element configured to change a travelling direction of light emitted from the light source.

13. An illuminator comprising: a light source supporter; a second substrate disposed to correspond to the light source supporter; a light source disposed on a surface of the light source supporter; and a meta surface on a first surface of the second substrate, wherein the light source and the meta surface are disposed to correspond to each other, wherein the meta surface comprises a plurality of nano-structures and each of the plurality of nano-structures has dimension less than a wavelength of light incident on the first surface, wherein the meta surface comprises: a first meta surface comprising a plurality of first nano-structures; and a second meta surface comprising a plurality of second nano-structures, and wherein the second meta surface comprises a micro-meta lens array provided to emit a non-patterned light having uniform illumination.
